

# Master's Capstone Thesis Roadmap

---

## MISM — IT Organizational Management & Leadership (UAGC)

---

### 1. Understanding Your Deliverable

Your program requires a capstone project (ISM 670), not a traditional thesis.

Per UAGC's ISM 670 course description, you must:

- Integrate professional practices from the MISM core courses with IT Organizational Management & Leadership specialization concepts
- Draw from **real organizational scenarios** to practice major IT concepts
- Select, develop, and present a **significant technology implementation project** incorporating organizational management, leadership strategies, systems development, and business planning

This is good news—it gives you more flexibility than a pure research thesis. You can build a framework/prototype, propose an implementation plan, and support it with research. The academic rigor comes from grounding your project in peer-reviewed literature and using a defensible methodology.

---

### 2. Recommended Topic: Multi-AI Governance Framework for IT Organizational Leadership

#### Why This Topic Wins

All five AIs you consulted converged on the same core idea: your existing practice of coordinating multiple AI platforms is academically novel and directly aligned with your degree. Here's why it's the strongest choice:

| Criterion                                           | Fit                                                            |
|-----------------------------------------------------|----------------------------------------------------------------|
| Aligns with ISM 670 requirements                    | ✓ Technology implementation project with leadership strategies |
| Draws from real organizational scenarios            | ✓ Your own documented multi-AI workflows                       |
| Incorporates organizational management & leadership | ✓ Governance, change management, decision-making               |
| Incorporates systems development                    | ✓ Framework/architecture design                                |
| Incorporates business planning                      | ✓ Strategic adoption roadmap for organizations                 |

| Criterion                             | Fit                                                       |
|---------------------------------------|-----------------------------------------------------------|
| Timely and novel                      | ✓ Enterprise multi-AI governance is a 2026 priority topic |
| Supported by academic literature      | ✓ TAM, Design Science Research, AI governance frameworks  |
| Positions you for IT leadership roles | ✓ CIO/CTO-level strategic thinking                        |

## Proposed Title

**“Designing a Multi-AI Governance Framework for Enterprise IT Decision Support: A Design Science Approach”**

## Research Questions

1. What governance architecture is needed to effectively coordinate outputs from multiple AI platforms in organizational decision-making contexts?
2. How does a structured multi-AI validation workflow compare to single-AI usage in terms of output accuracy, bias detection, and decision confidence?
3. What organizational leadership strategies are required to adopt and sustain multi-AI governance practices in IT management environments?

## 3. Methodology: Design Science Research (DSR)

Design Science Research is the gold standard for IS capstone projects that involve building an artifact (framework, model, prototype). It’s widely accepted in the information systems discipline and fits perfectly with what UAGC expects.

### The Six Phases (Peppers et al., 2007)

| Phase                        | What You Do                           | Your Application                                                                                       |
|------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------|
| 1. Problem Identification    | Define the organizational problem     | Organizations using multiple AI tools lack governance structures for cross-validation and coordination |
| 2. Objectives for a Solution | Define what a solution should achieve | A framework that improves decision quality, reduces AI bias, and provides audit-ready documentation    |
| 3. Design & Development      | Build the artifact                    | Create the Multi-AI Governance Framework (architecture                                                 |

| Phase            | What You Do           | Your Application                                                                                                                      |
|------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|                  |                       | diagram, workflow model, validation protocols, role definitions)                                                                      |
| 4. Demonstration | Show it works         | Apply the framework to 2–3 realistic organizational scenarios (e.g., strategic planning task, risk assessment, vendor evaluation)     |
| 5. Evaluation    | Measure effectiveness | Compare single-AI vs. multi-AI outputs on accuracy, completeness, and bias detection; survey IT professionals on perceived usefulness |
| 6. Communication | Present findings      | Your capstone paper + presentation                                                                                                    |

### Supporting Theoretical Frameworks

- **Technology Acceptance Model (TAM)** — To measure perceived usefulness and ease of use of the governance framework among IT professionals
- **TOE Framework (Technology-Organization-Environment)** — To analyze organizational readiness factors for multi-AI adoption
- **NIST AI Risk Management Framework** — To ground your governance model in established standards
- **Kotter’s 8-Step Change Model** — To address the leadership/change management dimension of adopting multi-AI workflows

## 4. Framework Architecture (The Core Artifact)

Your capstone deliverable is the framework itself. Here’s a high-level structure:

### Layer 1: AI Platform Selection & Role Assignment

- Define which AI platforms handle which tasks (e.g., research, code review, strategy analysis)
- Establish criteria for platform selection based on task type

### Layer 2: Cross-Validation Protocol

- Define when and how outputs from one AI are verified by another
- Establish consensus thresholds (e.g., 3-of-5 agreement)

- Document adversarial critique procedures

### Layer 3: Governance & Oversight

- Human-in-the-loop decision points
- Escalation protocols for conflicting AI outputs
- Audit trail requirements and documentation standards

### Layer 4: Organizational Integration

- Role definitions (who manages the multi-AI workflow)
- Training and change management plan
- KPI dashboard for monitoring framework effectiveness

### Layer 5: Ethical & Risk Management

- Bias detection mechanisms
  - Data privacy controls across platforms
  - Compliance mapping (NIST AI RMF, EU AI Act awareness)
- 

## 5. Data Collection Strategy

Since this is a capstone (not a full doctoral dissertation), you have flexibility:

### Primary Data

- **Self-documented case study:** Your own multi-AI workflow logs, prompt strategies, and cross-validation results (anonymized as needed)
- **Scenario testing:** Run 3 identical decision-support tasks through (a) single AI and (b) your multi-AI governance framework, then compare outputs on quality metrics

### Secondary Data (Optional but Strengthening)

- **Survey of IT professionals** (10–20 respondents via LinkedIn or professional networks) using TAM constructs to evaluate the framework’s perceived usefulness and adoptability
- **Expert review:** Have 2–3 IT professionals or faculty review your framework and provide structured feedback

### Metrics to Measure

| Metric                       | How Measured                                                 |
|------------------------------|--------------------------------------------------------------|
| Output accuracy/completeness | Rubric-scored comparison of single-AI vs. multi-AI responses |
| Bias detection rate          | Count of biases/errors caught by cross-validation            |

| Metric                      | How Measured                                   |
|-----------------------------|------------------------------------------------|
|                             | that single-AI missed                          |
| Decision confidence         | Self-reported Likert scale (if using survey)   |
| Perceived usefulness (TAM)  | Survey instrument                              |
| Perceived ease of use (TAM) | Survey instrument                              |
| Time to decision            | Clock time for single-AI vs. multi-AI workflow |

## 6. Chapter Outline

| Chapter                         | Content                                                                                                                   | Approx. Pages       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1. Introduction                 | Problem statement, purpose, research questions, significance                                                              | 5–7                 |
| 2. Literature Review            | AI governance frameworks, multi-agent systems, TAM/TOE, organizational change management, IT leadership                   | 12–18               |
| 3. Methodology                  | Design Science Research process, data collection plan, evaluation criteria                                                | 8–10                |
| 4. Framework Design             | The Multi-AI Governance Framework (architecture, layers, workflows, role definitions) — <i>this is your core artifact</i> | 15–20               |
| 5. Demonstration & Evaluation   | Scenario testing results, comparison metrics, survey/expert feedback analysis                                             | 10–15               |
| 6. Discussion                   | Implications for IT leadership, limitations, organizational adoption considerations                                       | 5–8                 |
| 7. Conclusion & Recommendations | Summary of findings, strategic adoption roadmap for organizations, future research                                        | 3–5                 |
| References                      | 40–60 sources                                                                                                             | 3–5                 |
| Appendices                      | Survey instrument, scenario prompts, raw data tables, framework diagrams                                                  | As needed           |
| <b>Total</b>                    |                                                                                                                           | <b>~60–90 pages</b> |

## 7. Key Literature to Start With

### AI Governance

- NIST AI Risk Management Framework (AI 100-1, 2023)
- EU AI Act overview documents
- Alrehili & Alsehli (2025), “The Transformative Role of AI in Organizational Decision-Making”
- Enterprise AI governance best practices (2026 industry reports)

### Multi-AI / Multi-Agent Systems

- Shu et al. (2024), “Towards Effective GenAI Multi-Agent Collaboration” (AWS/arXiv)
- HAIA-RECCLIN framework (Puglisi, 2025) — governance for multi-AI workflows
- Anthropic’s multi-agent research system architecture (2025)

### Technology Acceptance & Adoption

- Davis (1989), Technology Acceptance Model (TAM) — foundational
- Venkatesh et al. (2003), UTAUT — extended acceptance model
- Recent TAM + AI studies (2024–2025) in healthcare, education, enterprise

### Design Science Research

- Peffers et al. (2007), “A Design Science Research Methodology for Information Systems”
- Hevner et al. (2004), “Design Science in Information Systems Research”

### Organizational Change & IT Leadership

- Kotter (1996), “Leading Change”
  - Frank (2025), “Understanding Resistance to Organizational AI Adoption”
  - Impact of organizational culture on AI adoption (DSU dissertation, 2025)
- 

## 8. Timeline (Adjust to Your Course Schedule)

| Phase               | Tasks                                                                              | Weeks                    |
|---------------------|------------------------------------------------------------------------------------|--------------------------|
| <b>Pre-Capstone</b> | Finalize topic, begin literature review, draft problem statement                   | 4–6 weeks before ISM 670 |
| <b>Week 1–2</b>     | Submit proposal/problem statement, finalize research questions, set up methodology | 2 weeks                  |

| Phase             | Tasks                                                               | Weeks   |
|-------------------|---------------------------------------------------------------------|---------|
| <b>Week 3–4</b>   | Complete literature review, begin framework design                  | 2 weeks |
| <b>Week 5–6</b>   | Finish framework artifact, run scenario demonstrations              | 2 weeks |
| <b>Week 7–8</b>   | Collect evaluation data (metrics, survey/feedback), analyze results | 2 weeks |
| <b>Week 9–10</b>  | Write discussion, conclusion, finalize paper                        | 2 weeks |
| <b>Week 11–12</b> | Revise based on instructor feedback, prepare presentation           | 2 weeks |

*Note: UAGC courses typically run 5–6 weeks. Adjust this timeline to match your actual ISM 670 course length. Start the literature review and framework sketch BEFORE the capstone course begins.*

## 9. How This Connects to Your Current AI Work

| Your Current Practice                                             | Academic Translation                                                                      |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Using ChatGPT, Claude, Gemini, Grok, Copilot, Perplexity together | Multi-AI orchestration architecture                                                       |
| Cross-verifying outputs across platforms                          | Cross-validation governance protocol                                                      |
| Structured prompt workflows and documentation                     | Knowledge management and audit trail design                                               |
| Adversarial critique between AIs                                  | Bias detection and quality assurance mechanism                                            |
| Long-term planning and governance concepts                        | Strategic IT governance modeling                                                          |
| Omniversal Architect Labs organizational design                   | Conceptual prototype / simulation environment (cited as inspiration, not submitted as-is) |

**Critical rule:** Frame everything in professional/academic language. Your creative worldbuilding is the *inspiration and testbed*; the capstone deliverable is the *governance framework and its evaluation*.

## 10. Career Positioning

This capstone positions you for:

- **IT Governance roles** — You'll have a published framework for AI oversight
- **CIO/CTO advisory track** — Multi-AI strategy is a C-suite concern in 2026
- **AI Program Management** — Organizations are hiring for exactly this capability

- **Potential publication** — If your evaluation data is strong, this could become a conference paper or journal article
  - **Doctoral pathway** — This is structured enough to serve as a pilot study for a future PhD dissertation
- 

## Next Steps

1. **Confirm with your UAGC advisor** that this topic and methodology are acceptable for ISM 670
2. **Start your literature review now** — focus on the 5 areas listed in Section 7
3. **Begin documenting your current multi-AI workflows** in a structured format (this becomes your case study data)
4. **Draft a 1-page problem statement** and share it with your advisor and with your AI collaborators for refinement